Pre-trained models

A pre-trained model is a neural network whose weights and biases have already been learned from a large dataset and are made available for others to use.
Instead of training from scratch,these models can be used through transfer learning or fine-tuning. Here are the major pre-trained models categorized by application.

S.No	Category	Examples of Pre-trained Models	Main Applications
1	Image Classification (CNNs)	LeNet, AlexNet, VGG, ResNet, DenseNet, MobileNet, EfficientNet, ConvNeXt, NASNet, RegNet	Image classification
2	Object Detection	R-CNN, Fast R-CNN, Faster R-CNN, YOLO, SSD, RetinaNet, EfficientDet, DETR, RT-DETR	Object detection
3	Image Segmentation	FCN, U-Net, DeepLabV3+, PSPNet, SegFormer, Mask R-CNN, SAM	Semantic and instance segmentation
4	Vision Transformers (ViT)	ViT, DeiT, Swin Transformer, BEiT, DINOv2	Image classification, detection
5	Natural Language Processing (NLP)	BERT, RoBERTa, DistilBERT, ALBERT, XLNet, T5, BART	Text classification, QA, summarization
6	Large Language Models (LLMs)	GPT, Llama, Gemma, Mistral, Claude, Falcon, BLOOM, Qwen, Phi	Chatbots, coding, reasoning
7	Speech Models	DeepSpeech, Whisper, Wav2Vec 2.0, HuBERT, WavLM, SpeechT5	Speech recognition
8	Text-to-Speech (TTS)	VITS, FastSpeech2, Tacotron2, Bark	Voice synthesis
9	Audio Models	CLAP, AudioMAE, MusicGen, EnCodec	Audio classification, music generation
10	Generative Models	GAN, StyleGAN, Stable Diffusion, DALL·E, Imagen	Image generation
11	Video Generation Models	Sora, Gen-2, Pika, CogVideoX	Text-to-video generation
12	Multimodal Models	CLIP, BLIP, Flamingo, Kosmos-2, PaLI, Gemini	Image + text understanding
13	Recommendation Models	DeepFM, Wide & Deep, DLRM, DSSM	Recommendation systems

14	Time-Series Models	TimeGPT, PatchTST, TimesFM, Chronos	Forecasting
15	Graph Neural Networks (GNNs)	GCN, GraphSAGE, GAT, Graphormer	Graph analytics, social networks
16	Medical AI Models	MedSAM, BioBERT, ClinicalBERT, MedPaLM	Healthcare and medical imaging
17	Protein/Biology Models	AlphaFold, ESM-2, ProtBERT, OpenFold	Protein structure prediction
18	Reinforcement Learning Models	Decision Transformer, DreamerV3, MuZero	Robotics, game AI
19	OCR & Document AI	LayoutLM, Donut, TrOCR, PaddleOCR	Document understanding
20	Anomaly Detection Models	PaDiM, PatchCore, FastFlow	Industrial defect detection
21	Pose Estimation Models	OpenPose, HRNet, MediaPipe Pose, MoveNet	Human pose estimation
22	Face Recognition Models	FaceNet, ArcFace, InsightFace	Face recognition and verification
23	Image Captioning Models	BLIP, BLIP-2, OFA, GIT	Image-to-text generation
24	Visual Question Answering (VQA)	BLIP-2, Flamingo, LLaVA	Image-based question answering
25	Remote Sensing Models	Prithvi, SatMAE, GeoCLIP	Satellite imagery analysis

1. Image Classification (CNN Models)

An image classification pre-trained model is a CNN or Vision Transformer that has already learned to identify objects in images and can be reused for new image classification tasks.

S.No	Model	Year	Key Feature	Common Use	Link	
1	LeNet-5	1998	First successful CNN	Digit recognition	LeNet Paper	
2	AlexNet	2012	Started deep learning revolution	Image classification	AlexNet	
3	ZFNet	2013	Improved AlexNet	Image recognition	ZFNet paper	
4	VGG16	2014	Simple architecture with 16 layers	Transfer learning	VGG paper	
5	VGG19	2014	Deeper VGG	Classification		
6	GoogLeNet (Inception v1)	2014	Inception modules	Efficient classification	GoogLeNet paper	
7	InceptionV3	2015	Improved Inception	Image classification	InceptionV3 paper	
8	ResNet50	2015	Residual connections	Very popular transfer learning	ResNet paper,	
9	ResNet101	2015	Deeper ResNet	Feature extraction	torchvision	
10	ResNet152	2015	Very deep residual network	Research		
11	DenseNet121	2017	Dense connections	Medical imaging	DenseNet paper	
12	DenseNet169	2017	Improved DenseNet	Classification		
13	DenseNet201	2017	Deeper DenseNet	Feature extraction		

14	Xception	2017	Depthwise separable convolutions	Image classification	Xception paper	
15	MobileNetV1	2017	Lightweight	Mobile devices	MobileNetV1 paper	
16	MobileNetV2	2018	Inverted residuals	Embedded systems	MobileNetV2 paper	
17	MobileNetV3	2019	Optimized for mobile	Smartphones	MobileNetV3 paper	
18	EfficientNetB0–B7	2019	Excellent accuracy with fewer parameters	Transfer learning	EfficientNet paper	timm
19	ConvNeXt	2022	Modern CNN inspired by Transformers	Classification	ConvNeXt paper	Hugging Face
20	ViT (Vision Transformer)		Transformer-based image classifier		Hugging Face	
21	Swin Transformer		Hierarchical vision transformer		Hugging Face	
22	CLIP		Image-text joint embedding, zero-shot classification		OpenAI GitHub	
23	DINOv2		Self-supervised ViT, strong general features		Meta AI GitHub	

2. Object Detection Models

An Object Detection Pre-trained Model is a deep learning model that has already been trained to locate (detect) and identify multiple objects in an image or video using a large annotated dataset such as COCO or Open Images.
Unlike image classification, object detection tells you what the objects are and where they are.

S.No	Model	Year	Main Application	Advantages	Official Link	Type
1	R-CNN	2014	Object detection in images	High accuracy, first successful CNN detector	R-CNN Paper	Two-stage
2	Fast R-CNN	2015	Object detection	Faster than R-CNN by sharing convolutional features	Fast R-CNN Paper	Two-stage

3	Faster R-CNN	2015	Object detection, face detection, medical imaging	Introduced Region Proposal Network (RPN), excellent accuracy	Faster R-CNN Paper	Two-stage
4	Mask R-CNN	2017	Instance segmentation, autonomous driving, medical image analysis	Detects objects and generates segmentation masks	Mask R-CNN Paper	Two-stage
5	YOLO (You Only Look Once)	2016–Present	Real-time object detection, surveillance, drones, robotics, traffic monitoring	Very fast, suitable for real-time applications	Ultralytics YOLO Documentation	One-stage
6	SSD (Single Shot Detector)	2016	Mobile applications, embedded systems, autonomous vehicles	Good balance between speed and accuracy	SSD Paper	One-stage
7	RetinaNet	2017	Object detection in dense scenes	Uses Focal Loss to improve detection of rare objects	RetinaNet Paper	One-stage
8	EfficientDet	2020	Industrial inspection, autonomous driving	High accuracy with fewer parameters	EfficientDet Paper	One-stage
9	DETR (Detection Transformer)	2020	Object detection	End-to-end detection without anchor boxes or NMS	DETR Paper	Transformer
10	Deformable DETR	2021	Small object detection	Faster convergence and improved small-object performance	Deformable DETR Paper	Transformer
11	YOLOS	2021	General object detection	Pure Vision Transformer detector	YOLOS Paper	Vision Transformer
12	RT-DETR	2023	Real-time object detection	Combines transformer accuracy with real-time speed	RT-DETR Paper	Transformer

3. Image Segmentation Models

A pre-trained image segmentation model is a segmentation architecture whose weights have already been trained on large datasets such as ImageNet, COCO, ADE20K, Cityscapes, or Pascal VOC. Instead of training from scratch, you can fine-tune these models on your own dataset. Many are available through TensorFlow/Keras, PyTorch, and Hugging Face.

S.No	Pre-trained Model	Backbone	Pre-trained Dataset	Applications	Official Link	Segmentation Type
1	FCN-ResNet50	ResNet-50	COCO / ImageNet	Road scenes, satellite images	TorchVision FCN Documentation	Semantic
2	FCN-ResNet101	ResNet-101	COCO / ImageNet	General semantic segmentation	TorchVision FCN Documentation	Semantic
3	DeepLabV3-ResNet50	ResNet-50	COCO / ImageNet	Medical imaging, autonomous driving	TorchVision DeepLabV3 Documentation	Semantic
4	DeepLabV3-ResNet101	ResNet-101	COCO / ImageNet	Satellite imagery, scene understanding	TorchVision DeepLabV3 Documentation	Semantic
5	LRASPP-MobileNetV3	MobileNetV3	COCO	Mobile and embedded devices	TorchVision LRASPP Documentation	Semantic
6	U-Net (Pre-trained Encoder)	VGG16 / ResNet / EfficientNet	ImageNet	Medical image segmentation	Segmentation Models Library (Keras)	Semantic
7	UNet++	ResNet / EfficientNet	ImageNet	Tumor and organ segmentation	Segmentation Models Library (Keras)	Semantic
8	FPN (Feature Pyramid Network)	ResNet / EfficientNet	ImageNet	Remote sensing, aerial images	Segmentation Models Library (Keras)	Semantic
9	PSPNet	ResNet	ImageNet	Urban scene understanding	Segmentation Models Library (Keras)	Semantic
10	LinkNet	ResNet	ImageNet	Real-time segmentation	Segmentation Models Library (Keras)	Semantic
11	SegFormer-B0	MiT-B0 Transformer	ADE20K	General-purpose segmentation	Hugging Face SegFormer Models	Semantic
12	SegFormer-B1	MiT-B1 Transformer	ADE20K	Scene parsing	Hugging Face SegFormer Models	Semantic
13	SegFormer-B2	MiT-B2 Transformer	ADE20K	Medical and satellite images	Hugging Face SegFormer Models	Semantic

14	SegFormer-B5	MiT-B5 Transformer	ADE20K	High-accuracy segmentation	Hugging Face SegFormer Models	Semantic
15	Mask2Former	Swin Transformer	COCO / ADE20K	Semantic, instance, panoptic segmentation	Hugging Face Mask2Former Models	Universal
16	OneFormer	Swin Transformer	ADE20K / COCO	Multi-task segmentation	Hugging Face OneFormer Models	Universal
17	DETR Panoptic	ResNet-50	COCO	Autonomous driving	Hugging Face DETR Panoptic Models	Panoptic
18	Segment Anything Model (SAM)	Vision Transformer	SA-1B	Image editing, annotation, robotics	Segment Anything Project	Interactive

4. Natural Language Processing (NLP)

KerasHub provides ready-to-use pre-trained models that can be loaded using from_preset(), including BERT, ALBERT, BART, BLOOM, Gemma, GPT-2, Llama 3, Mistral, RoBERTa, T5, Whisper, and XLM-RoBERTa.

S.No	Pre-trained Model	Developed By	Main Applications	Common NLP Tasks	Official Link
1	**Word2Vec**	Google	Word embeddings	Similarity, text classification	Word2Vec Paper
2	**GloVe**	Stanford	Word embeddings	Semantic similarity, sentiment analysis	GloVe Project
3	**FastText**	Meta	Word and subword embeddings	Text classification, multilingual NLP	FastText Project
4	**ELMo**	Allen Institute for AI	Contextual word embeddings	Named Entity Recognition (NER), Question Answering	ELMo Paper
5	**BERT**	Google	Language understanding	Text classification, NER, QA, sentiment analysis	BERT Documentation (KerasHub)
6	**RoBERTa**	Meta	Improved BERT	Classification, NER, QA	RoBERTa Paper
7	**DistilBERT**	Hugging Face	Lightweight BERT	Mobile NLP, sentiment analysis	DistilBERT Paper

8	**ALBERT**	Google	Memory-efficient BERT	Classification, QA	ALBERT Paper
9	**XLNet**	Google & CMU	Autoregressive Transformer	Text classification, QA	XLNet Paper
10	**Transformer-XL**	Google	Long-context language modeling	Language modeling, text generation	Transformer-XL Paper
11	**XLM**	Meta	Cross-lingual NLP	Translation, multilingual classification	XLM Paper
12	**XLM-RoBERTa**	Meta	Multilingual language understanding	Translation, multilingual NER	XLM-R Paper
13	**T5 (Text-to-Text Transfer Transformer)**	Google	Text generation	Translation, summarization, QA	T5 Paper
14	**BART**	Meta	Seq2Seq Transformer	Summarization, translation, text generation	BART Paper
15	**PEGASUS**	Google	Abstractive summarization	Document summarization	PEGASUS Paper
16	**GPT**	OpenAI	Text generation	Chatbots, code generation, writing	OpenAI Research
17	**GPT-2**	OpenAI	Language generation	Story generation, text completion	GPT-2 Paper
18	**BLOOM**	BigScience	Multilingual LLM	Translation, text generation	BLOOM Project
19	**Llama**	Meta	Large Language Model	Chatbots, reasoning, coding	Llama Official Site
20	**Mistral**	Mistral AI	Efficient LLM	Chatbots, summarization, coding	Mistral AI
21	**Gemma**	Google	Lightweight LLM	Chatbots, reasoning, NLP	Gemma Documentation

5.Vision Transformers

These models are used for Automatic Speech Recognition (ASR), Text-to-Speech (TTS), Speech Translation, Speaker Recognition, Voice Conversion, and Speech Understanding.

S.No	Pre-trained Model	Developed By	Primary Application	Common Tasks	Official Link
1	**Wav2Vec 2.0**	Meta	Self-supervised speech representation	Speech-to-Text (ASR), speech classification	Wav2Vec 2.0 Paper
2	**HuBERT**	Meta	Self-supervised speech learning	ASR, speech understanding	HuBERT Paper
3	**WavLM**	Microsoft	Universal speech model	ASR, speaker verification, emotion recognition	WavLM Paper
4	**Whisper**	OpenAI	Multilingual speech recognition	Speech-to-Text, translation	Whisper Paper
5	**SpeechT5**	Microsoft	Unified speech/text Transformer	ASR, TTS, speech translation, voice conversion	SpeechT5 Paper
6	**SeamlessM4T**	Meta	Multimodal multilingual speech	Speech-to-Speech, Speech-to-Text, Text-to-Speech, Translation	SeamlessM4T Project
7	**Conformer**	Google	Convolution + Transformer ASR	Automatic Speech Recognition	Conformer Paper
8	**Emformer**	Meta	Streaming Transformer	Real-time ASR	Emformer Paper
9	**Bark**	Suno AI	Transformer-based Text-to-Audio	Text-to-Speech, music, sound effects	Bark Model
10	**Speech2Text (S2T)**	Meta	Speech translation	Speech-to-Text translation	Speech2Text Documentation
11	**MMS (Massively Multilingual Speech)**	Meta	Multilingual speech	ASR, TTS, language identification	MMS Project
12	**VITS**	Kakao Brain	End-to-end Text-to-Speech	Natural voice synthesis	VITS Paper
13	**FastSpeech 2**	Microsoft	Fast Text-to-Speech	Voice assistants, audiobook generation	FastSpeech 2 Paper
14	**Voice Transformer Network (VTN)**	Nagoya University	Voice conversion	Speaker voice conversion	Voice Transformer Network Paper

6. Speech Models

Speech models focus on understanding or generating speech (ASR, TTS, speaker recognition, etc.). Some are Transformer-based, while others combine CNNs, RNNs, or Conformers.

S.No	Pre-trained Speech Model	Developed By	Main Application	Common Tasks	Framework Support	Official Link
1	**DeepSpeech**	Mozilla	Automatic Speech Recognition (ASR)	Speech-to-Text	TensorFlow, PyTorch	DeepSpeech Paper
2	**Jasper**	NVIDIA	Speech Recognition	ASR	NVIDIA NeMo	NVIDIA NeMo Speech Models
3	**QuartzNet**	NVIDIA	Lightweight ASR	Speech-to-Text	NVIDIA NeMo	NVIDIA NeMo Speech Models
4	**Conformer**	Google	Speech Recognition	ASR	TensorFlow, PyTorch	Conformer Paper
5	**Wav2Vec 2.0**	Meta	Self-supervised Speech Learning	ASR, Speech Classification	Hugging Face, PyTorch	Wav2Vec 2.0 Paper
6	**HuBERT**	Meta	Speech Representation Learning	ASR, Speaker Recognition	Hugging Face	HuBERT Paper
7	**WavLM**	Microsoft	Universal Speech Model	ASR, Speaker Verification, Emotion Recognition	Hugging Face	WavLM Paper
8	**Whisper**	OpenAI	Multilingual Speech Recognition	Speech-to-Text, Translation	Hugging Face, OpenAI	Whisper Paper
9	**SpeechT5**	Microsoft	Unified Speech & Text Processing	ASR, TTS, Speech Translation, Voice Conversion	Hugging Face	SpeechT5 GitHub
10	**Speech2Text (S2T)**	Meta	Speech Translation	Speech-to-Text Translation	Hugging Face	Speech2Text Documentation
11	**MMS (Massively Multilingual Speech)**	Meta	Multilingual Speech Processing	ASR, TTS, Language Identification	Hugging Face	MMS Project
12	**SeamlessM4T**	Meta	Universal Speech Translation	Speech-to-Speech, Speech-to-Text, Text-to-Speech	Hugging Face	SeamlessM4T Project
13	**FastSpeech 2**	Microsoft	Text-to-Speech	Natural Voice Synthesis	TensorFlow, PyTorch	FastSpeech 2 Paper

14	**VITS**	Kakao Brain	End-to-End Text-to-Speech	Voice Synthesis	PyTorch	VITS Paper
15	**Bark**	Suno AI	Text-to-Audio	TTS, Music, Sound Effects	Hugging Face	Bark Model

7. Generative Models

A Generative Pre-trained Model is an AI model that has already been trained on a very large dataset and can generate new content such as text, images, audio, music, video, or code based on what it has learned.
Unlike traditional models that only recognize or classify data, generative models can create new content.

7.a - Pre-trained Generative Models

S.No	Pre-trained Model	Developed By	Model Type	Main Application	Common Tasks	Official Link
1	**GPT (Generative Pre-trained Transformer)**	OpenAI	Transformer (LLM)	Natural Language Generation	Chatbots, text generation, coding	OpenAI Research
2	**GPT-2**	OpenAI	Transformer	Text Generation	Story writing, text completion	GPT-2 Paper
3	**GPT-3**	OpenAI	Transformer	General-purpose LLM	Chatbots, summarization, translation	GPT-3 Paper
4	**Llama 2 / Llama 3**	Meta	Transformer	Open-source LLM	Chatbots, coding, reasoning	Llama Official Site
5	**Mistral 7B / Mixtral**	Mistral AI	Transformer (MoE)	Efficient LLM	Chatbots, code generation	Mistral AI
6	**Gemma**	Google	Transformer	Lightweight LLM	NLP, chatbots, reasoning	Gemma Documentation
7	**BLOOM**	BigScience	Transformer	Multilingual Text Generation	Translation, generation	BLOOM Project
8	**Falcon**	Technology Innovation Institute	Transformer	Large Language Model	Text generation, coding	Falcon Models
9	**Claude Models**	Anthropic	Transformer	Conversational AI	Writing, reasoning, coding	Anthropic
10	**PaLM 2**	Google	Transformer	Language Understanding	Translation, coding	Google AI Research

7.b - Image Generation Models					
S.No	**Model**	**Developed By**	**Application**	**Tasks**	**Official Link**
1	**DALL·E 2 / DALL·E 3**	OpenAI	Text-to-Image	Image generation	**OpenAI Research**
2	**Stable Diffusion**	Stability AI	Text-to-Image	Image generation, editing	**Stability AI Open Models**
3	**Stable Diffusion XL (SDXL)**	Stability AI	Image Generation	High-resolution image synthesis	**Stability AI Open Models**
4	**Imagen**	Google	Text-to-Image	Photorealistic image generation	**Google Research Publications**
5	**Parti**	Google	Text-to-Image	High-quality image generation	**Google Research Publications**
6	**Midjourney**	Midjourney	AI Art	Creative image generation	**Midjourney**

7.c - Diffusion Models				
S.No	**Model**	**Developed By**	**Application**	**Official Link**
1	DDPM (Denoising Diffusion Probabilistic Models)	OpenAI	Image generation	DDPM Paper
2	Latent Diffusion Model (LDM)	LMU Munich & Stability AI	Efficient image generation	LDM Paper
3	Stable Diffusion	Stability AI	Text-to-image generation	Stable Diffusion Paper

7.d - GAN (Generative Adversarial Network) Models				
S.No	**Model**	**Developed By**	**Application**	**Official Link**
1	**GAN**	Ian Goodfellow et al.	Image generation	GAN Paper
2	**DCGAN**	Facebook AI	Image synthesis	DCGAN Paper
3	**CycleGAN**	UC Berkeley	Image-to-image translation	CycleGAN Project
4	**Pix2Pix**	UC Berkeley	Image translation	Pix2Pix Project
5	**StyleGAN**	NVIDIA	Face generation	StyleGAN Project
6	**StyleGAN2**	NVIDIA	High-quality face synthesis	StyleGAN2 Project
7	**StyleGAN3**	NVIDIA	Advanced image synthesis	StyleGAN3 Project
8	**BigGAN**	DeepMind	High-resolution image generation	BigGAN Paper

7.e - Speech & Audio Generative Models				
S.No	**Model**	**Developed By**	**Application**	**Official Link**
1	**Whisper**	OpenAI	Speech recognition & translation	Whisper Paper
2	**Bark**	Suno AI	Text-to-Speech	Bark Model
3	**VITS**	Kakao Brain	Natural Text-to-Speech	VITS Paper
4	**MusicGen**	Meta	Music generation	MusicGen Project

7.f - Video Generation Models				
S.No	**Model**	**Developed By**	**Application**	**Official Link**
1	**Sora**	OpenAI	Text-to-Video	Sora Research
2	**Gen-2**	Runway	Video generation	Runway Research
3	**Pika**	Pika Labs	AI video creation	Pika

Applications and recommended models		
S.No	**Application**	**Recommended Models**
1	Chatbots	GPT, Llama, Claude, Gemma
2	Code Generation	GPT, Llama, Mistral
3	Text Summarization	GPT, Claude, PaLM 2
4	Image Generation	Stable Diffusion, DALL·E, Midjourney, Imagen
5	Image Editing	Stable Diffusion, DALL·E
6	Face Generation	StyleGAN2, StyleGAN3
7	Music Generation	MusicGen
8	Text-to-Speech	Bark, VITS
9	Speech Recognition	Whisper
10	Video Generation	Sora, Gen-2, Pika
Popular repositories for pre-trained generative models		
1	Hugging Face Models	
2	KerasHub Pretrained Models	
3	NVIDIA NGC Model Catalog	